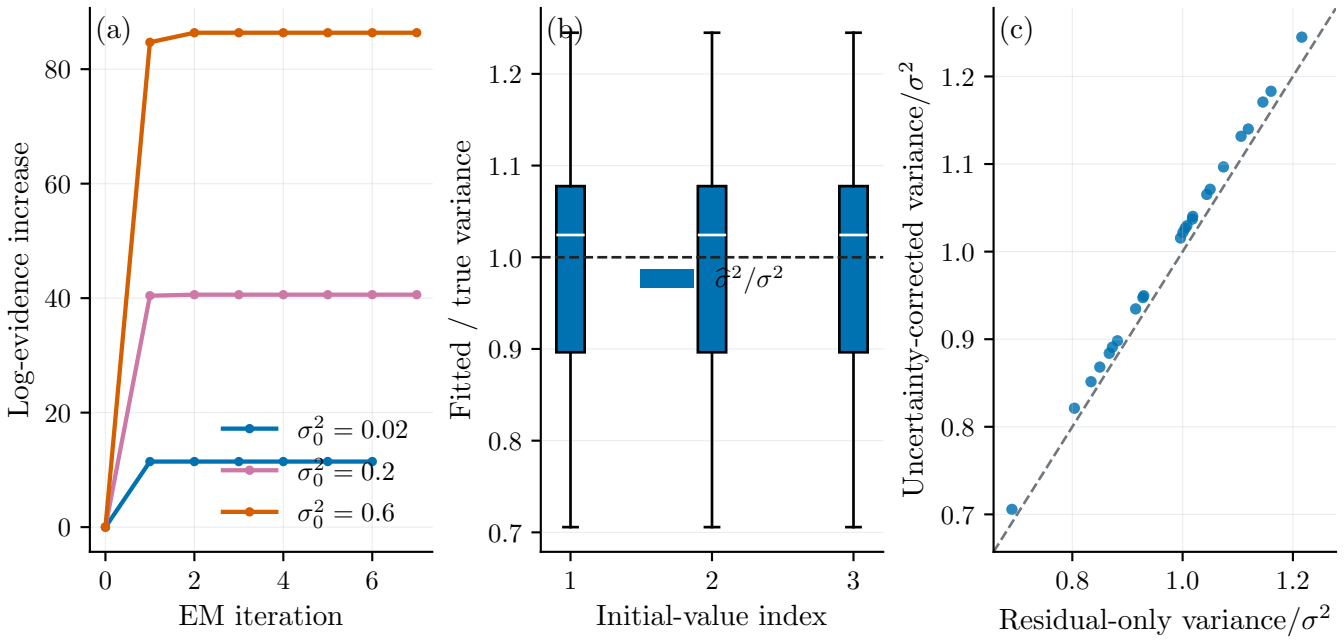
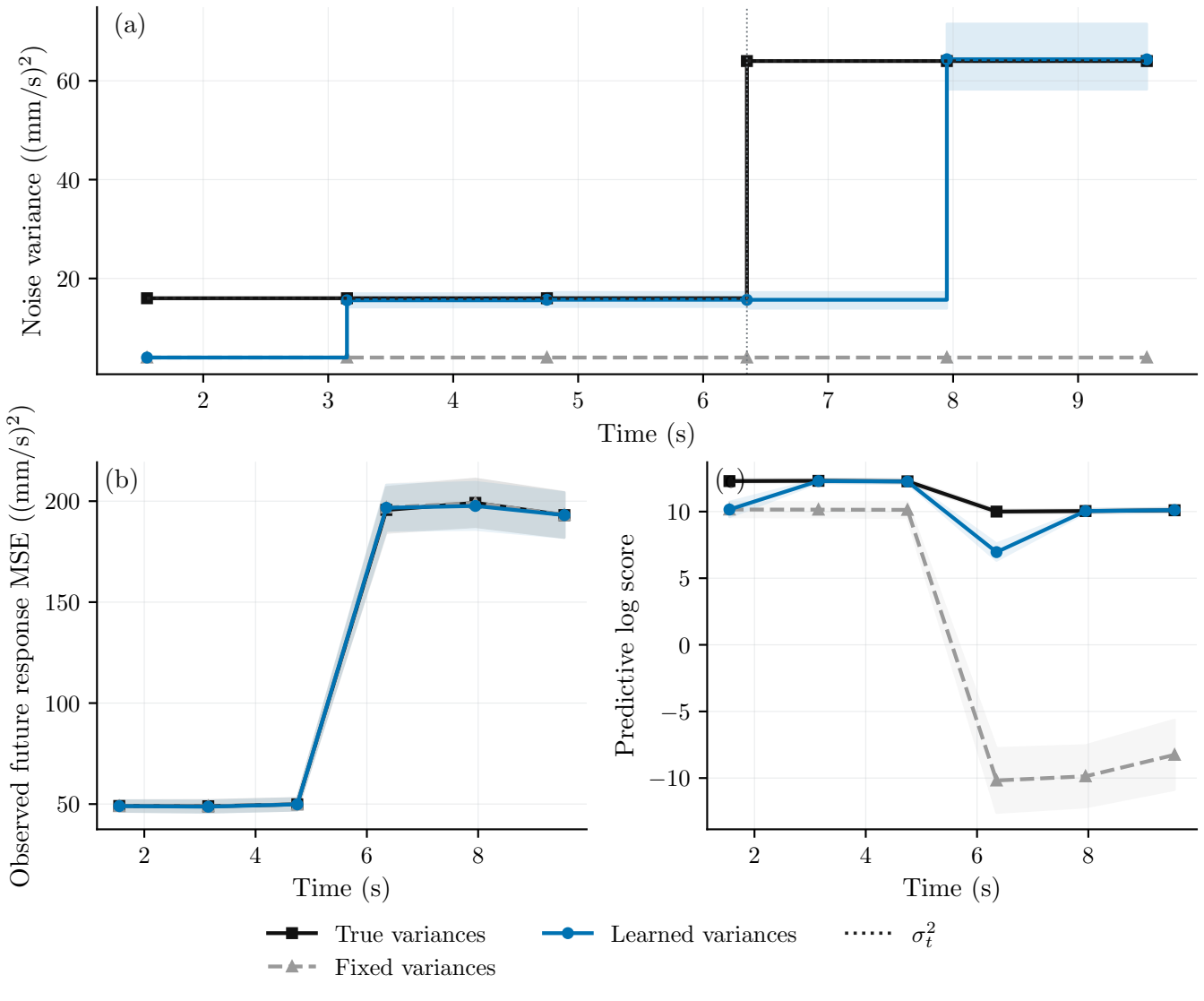
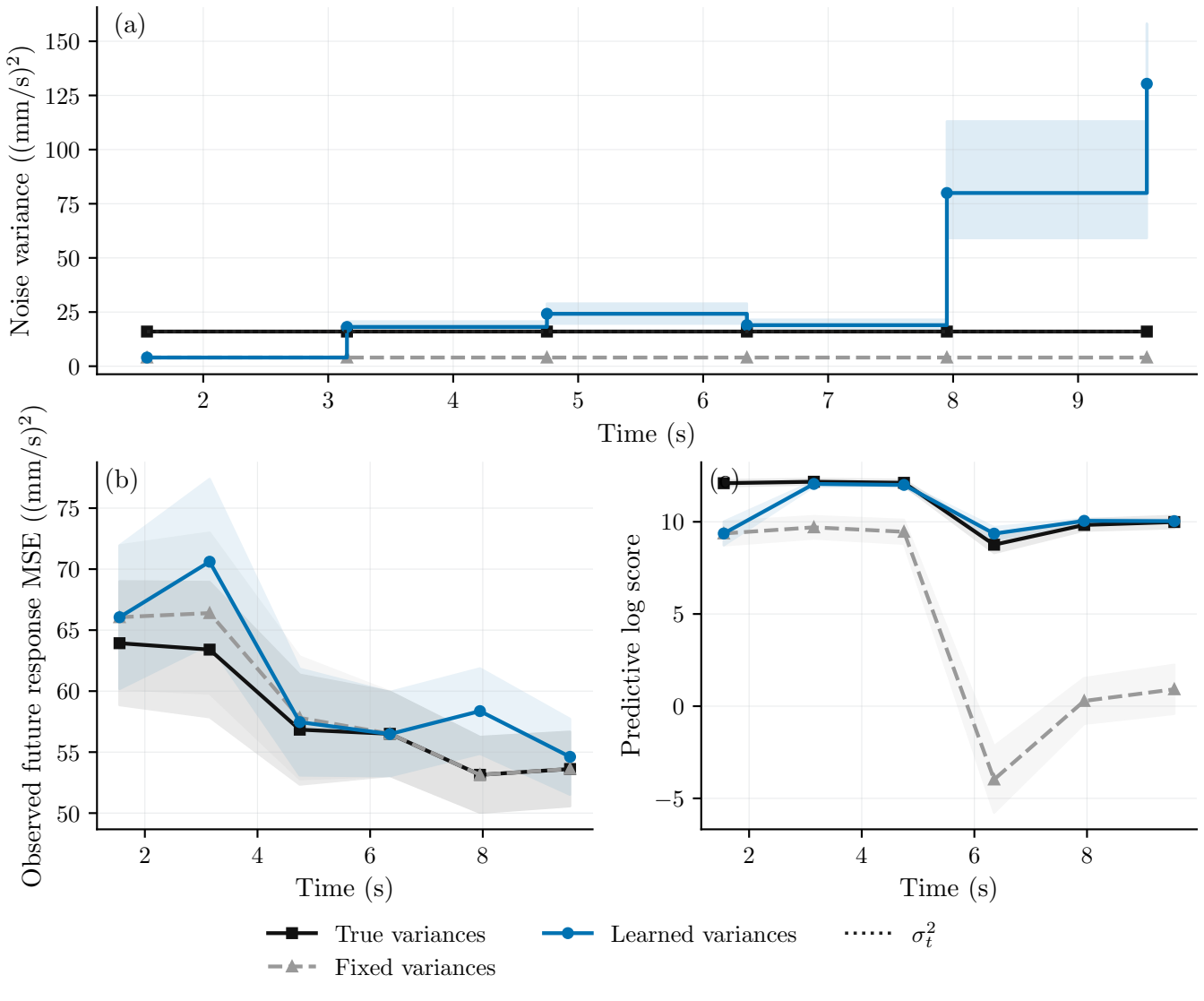
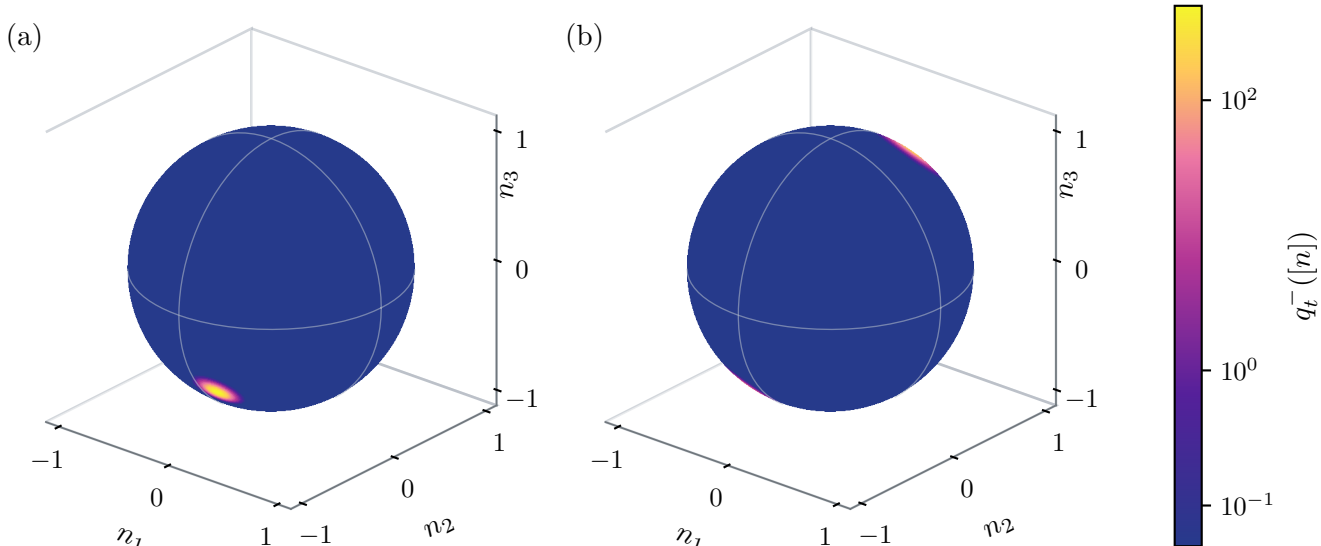
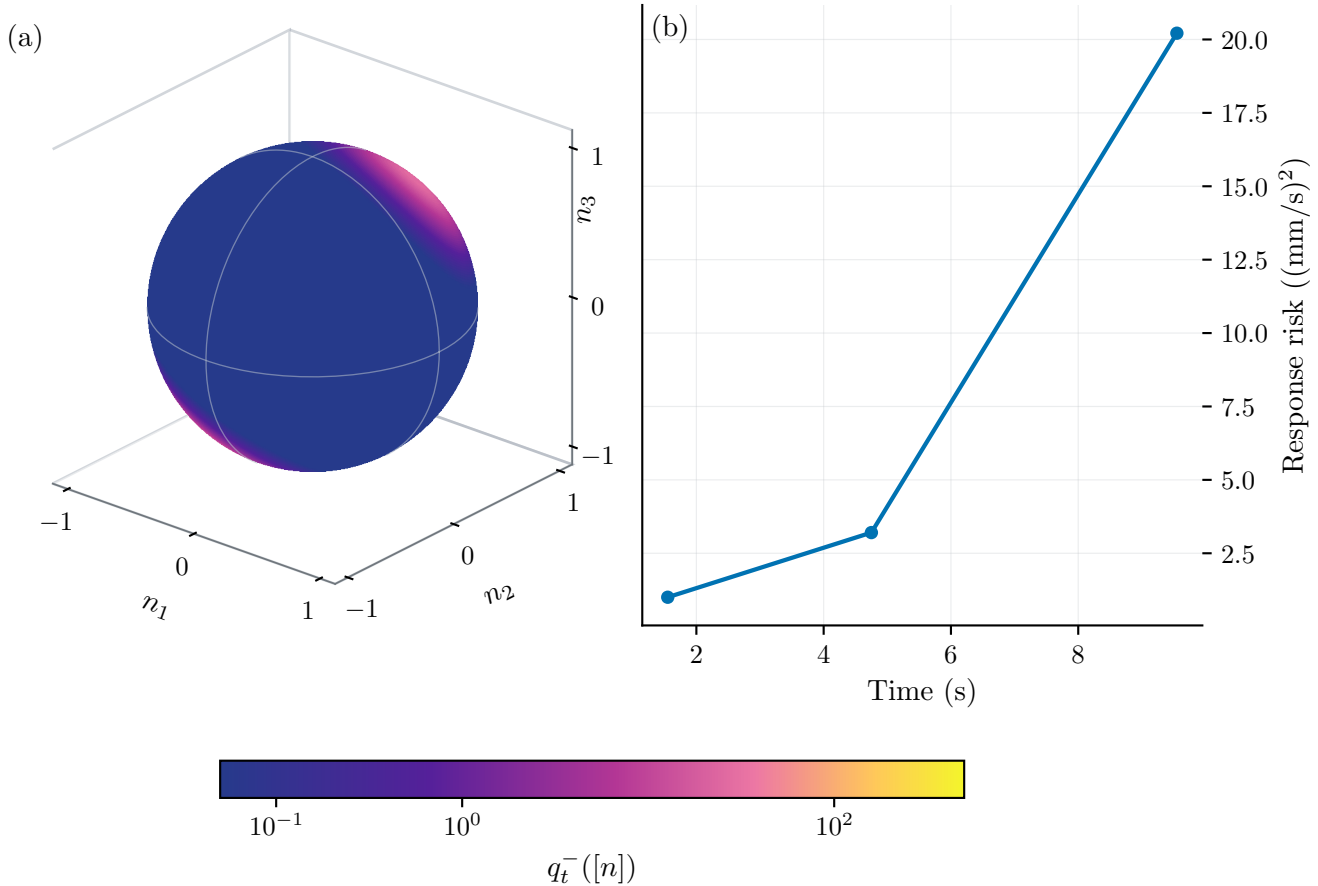
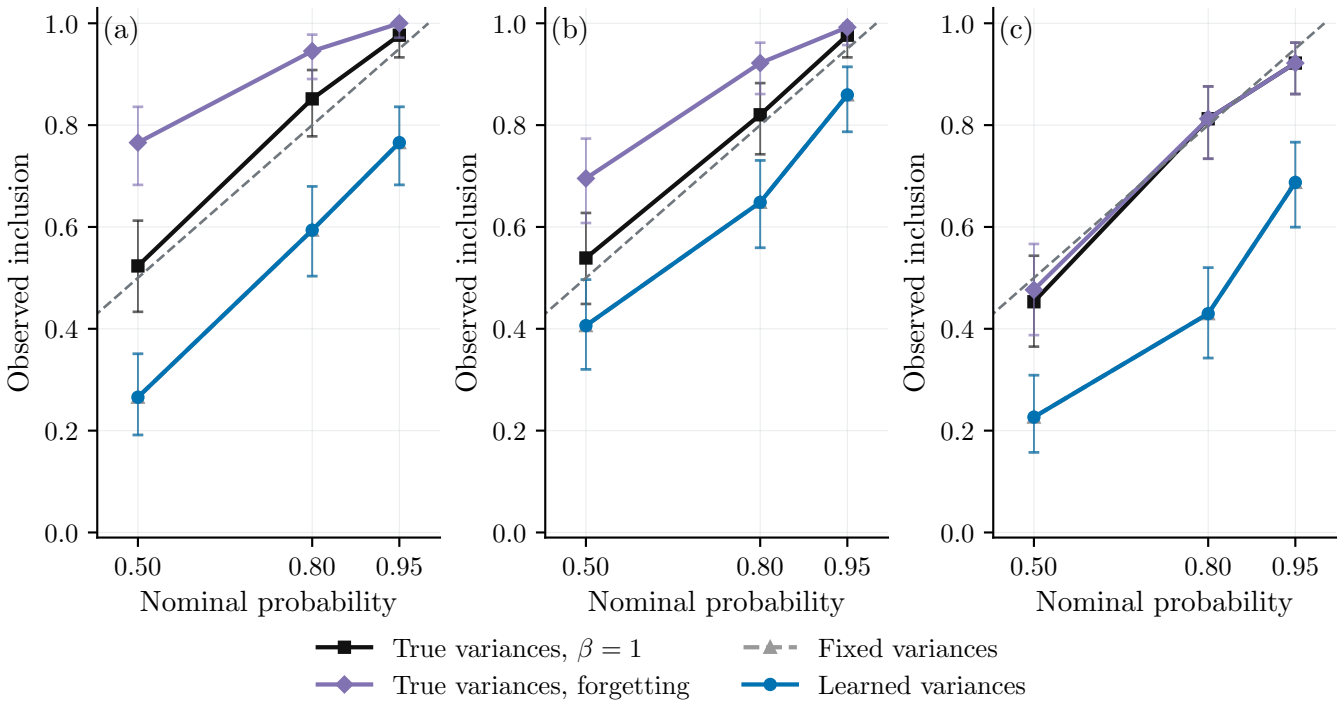

(a)
Log-evidence increase
EM iteration
$\sigma_0^2 = 0.02$
$\sigma_0^2 = 0.2$
$\sigma_0^2 = 0.6$
(b)
Fitted / true variance
Initial-value index
$\widehat{\sigma}^2/\sigma^2$
(c)
Uncertainty-corrected variance/σ^2
Residual-only variance/σ^2

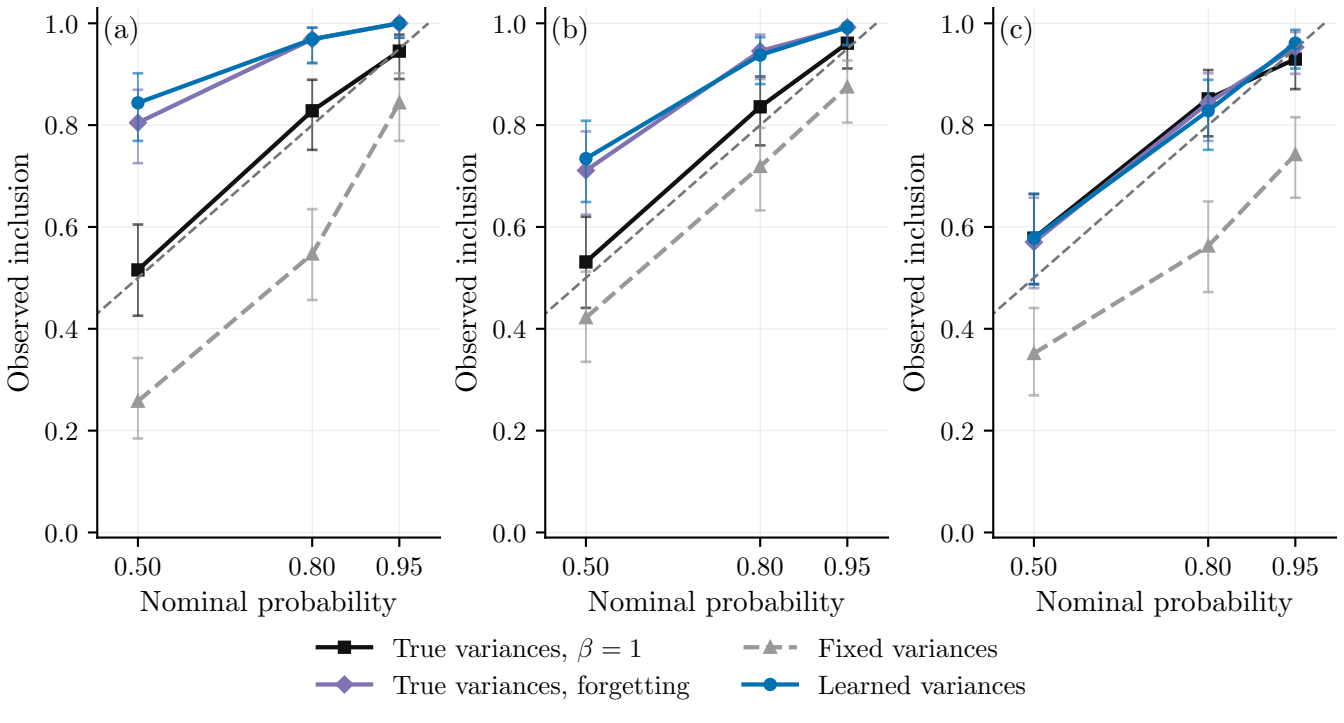

(a)
Noise variance ((mm/s)^2)
60
40
20
2
3
4
5
6
7
8
9
Time (s)
(b)
Observed future response MSE ((mm/s)^2)
200
150
100
50
2
4
6
8
Time (s)
(c)
Predictive log score
10
5
0
−5
−10
2
4
6
8
Time (s)
True variances
Learned variances
σ_t^2
Fixed variances

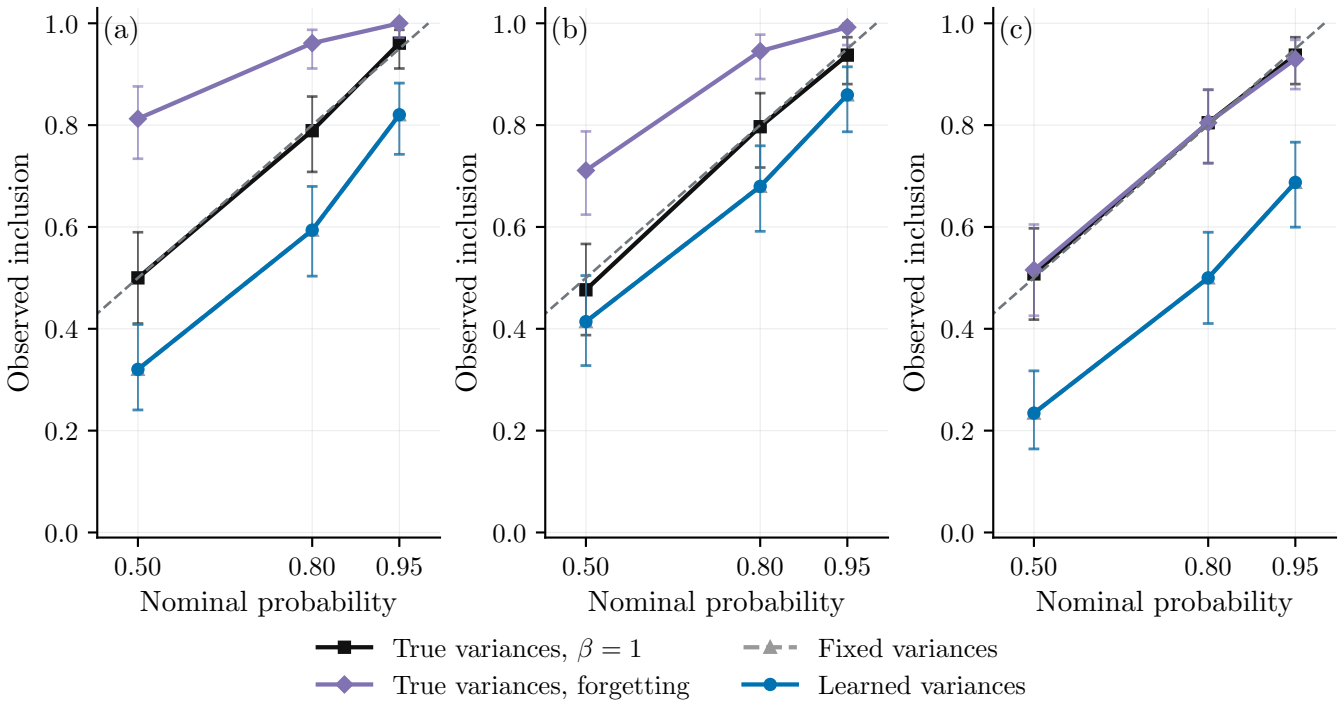

(a)
Noise variance ((mm/s)^2)
Time (s)
(b)
Observed future response MSE ((mm/s)^2)
Time (s)
(c)
Predictive log score
Time (s)
True variances
Learned variances
σ_t^2
Fixed variances

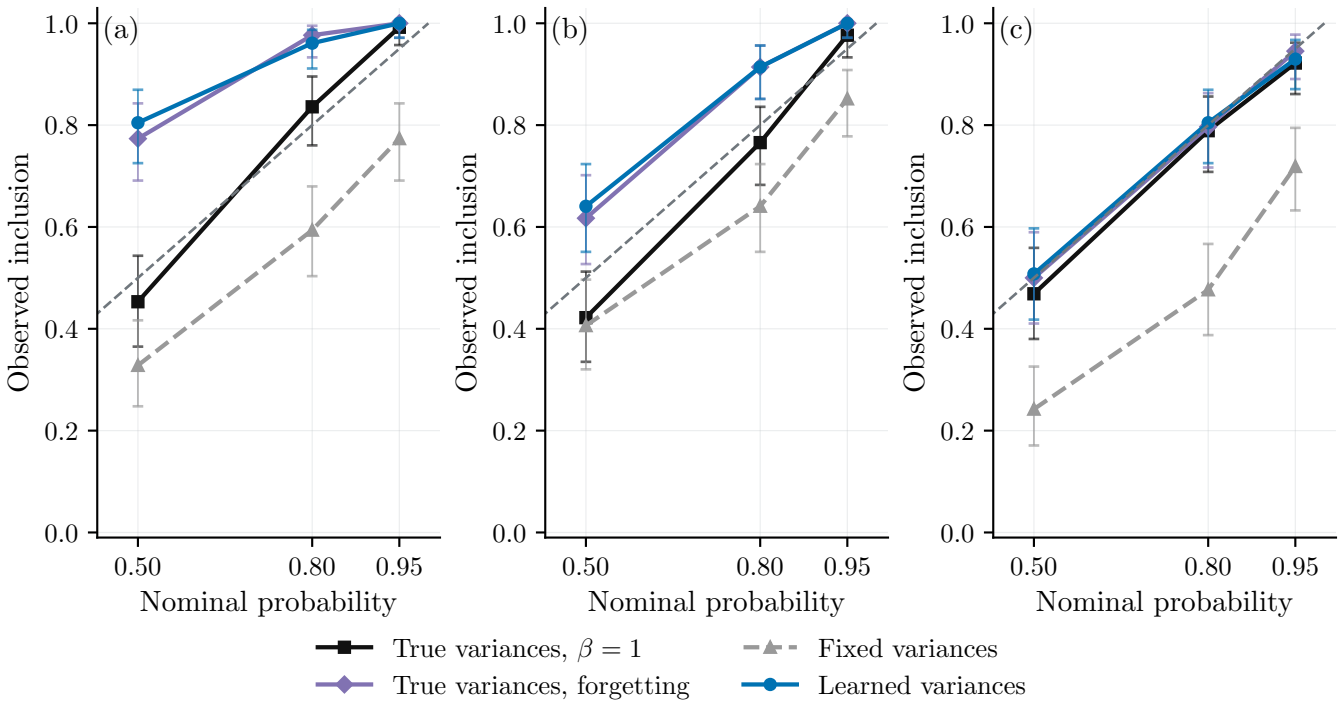

(a)
(b)
n_3
1
0
−1
n_1
n_2
$q_{-b}(|n|)$
10^2
10^0
10^{-1}

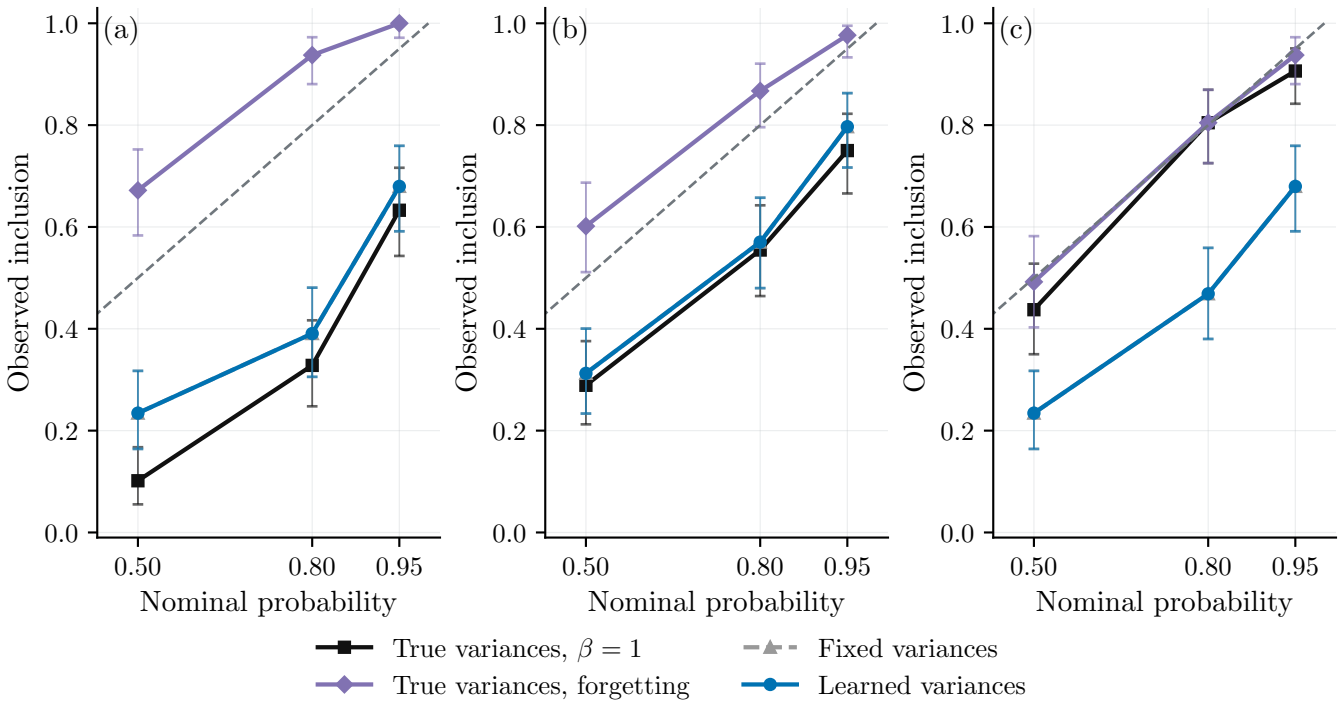

(a)
n_1
n_2
n_3
−1
0
1
(b)
Time (s)
Response risk ((mm/s)^2)
2
4
6
8
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
10^{-1}
10^{0}
10^{2}
$q_t^-([n])$

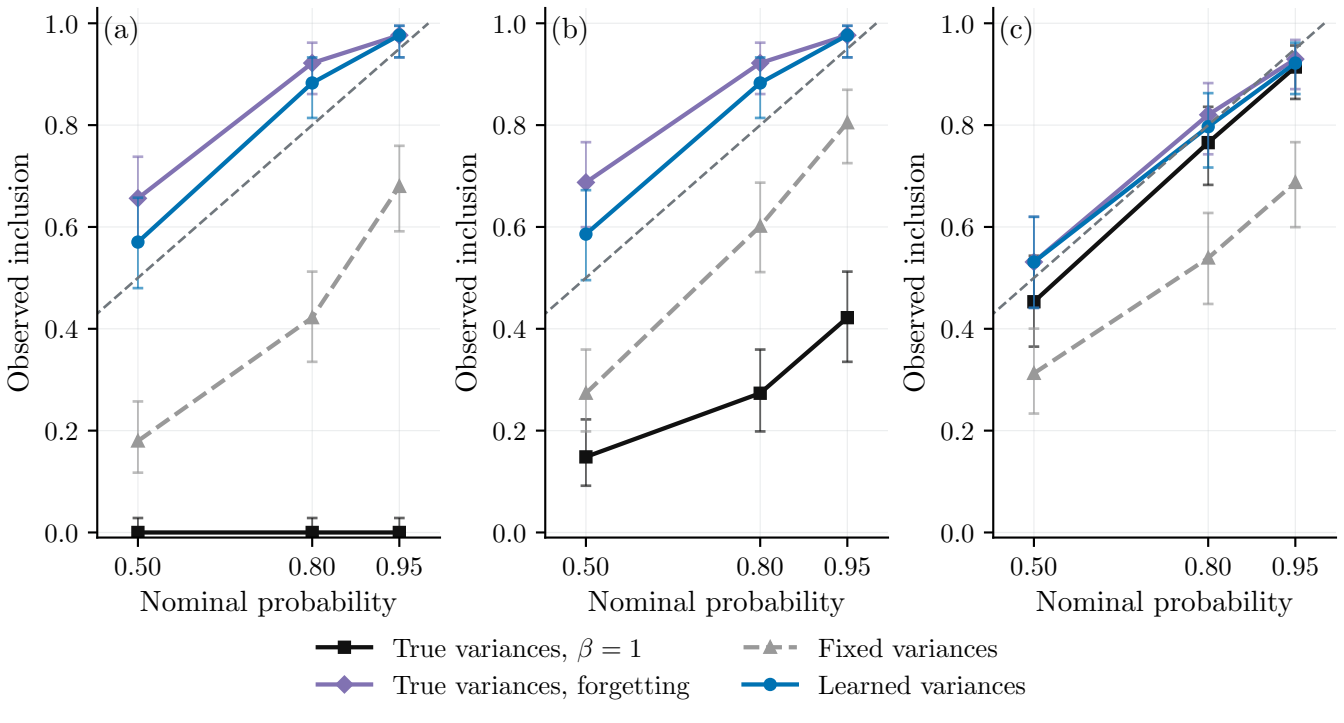

(a)
(b)
(c)
Observed inclusion
Nominal probability
0.0
0.2
0.4
0.6
0.8
1.0
0.50
0.80
0.95
True variances, $\beta = 1$
True variances, forgetting
Fixed variances
Learned variances

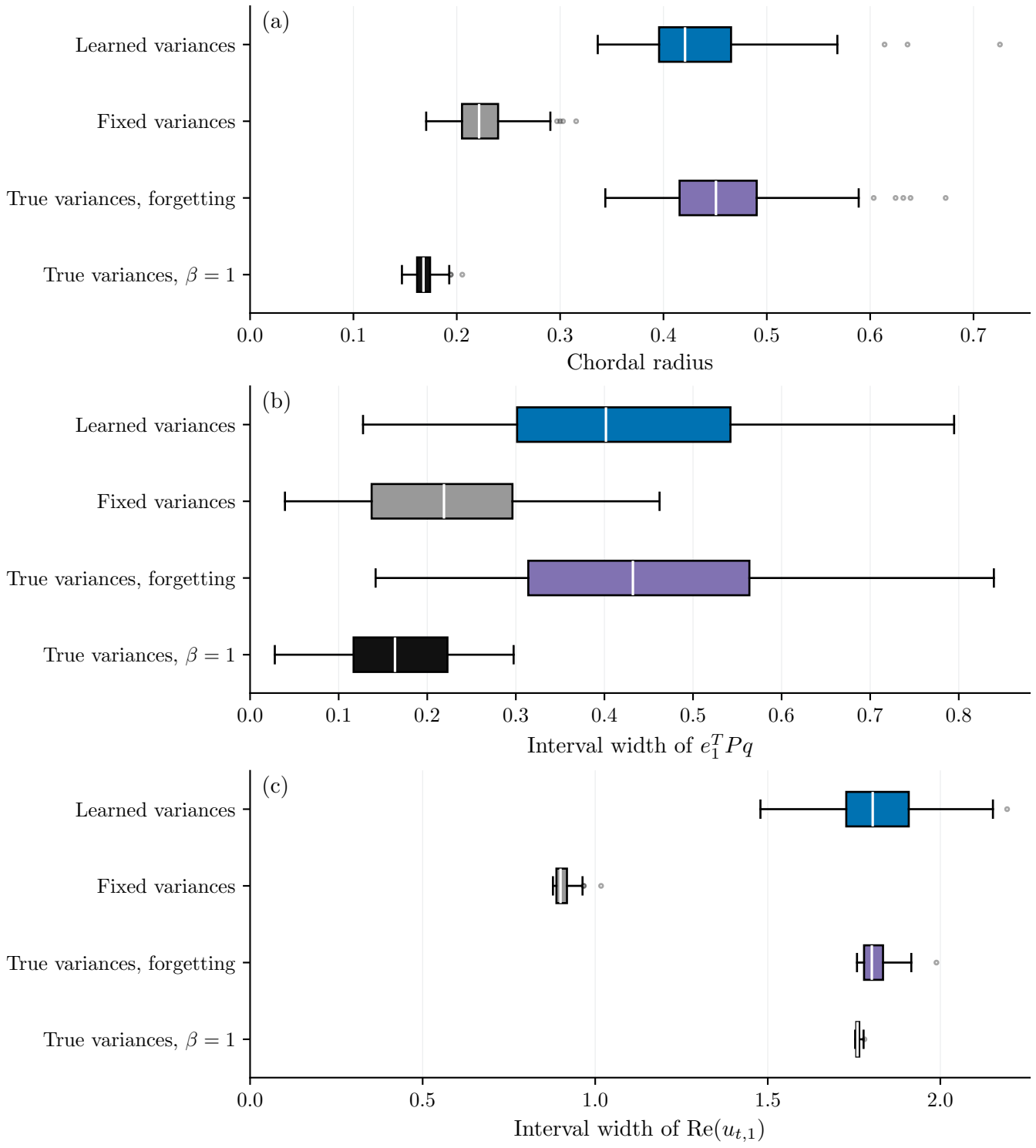

(a)
(b)
(c)
Observed inclusion
1.0
0.8
0.6
0.4
0.2
0.0
0.50
0.80
0.95
Nominal probability
True variances, $\beta = 1$
True variances, forgetting
Fixed variances
Learned variances

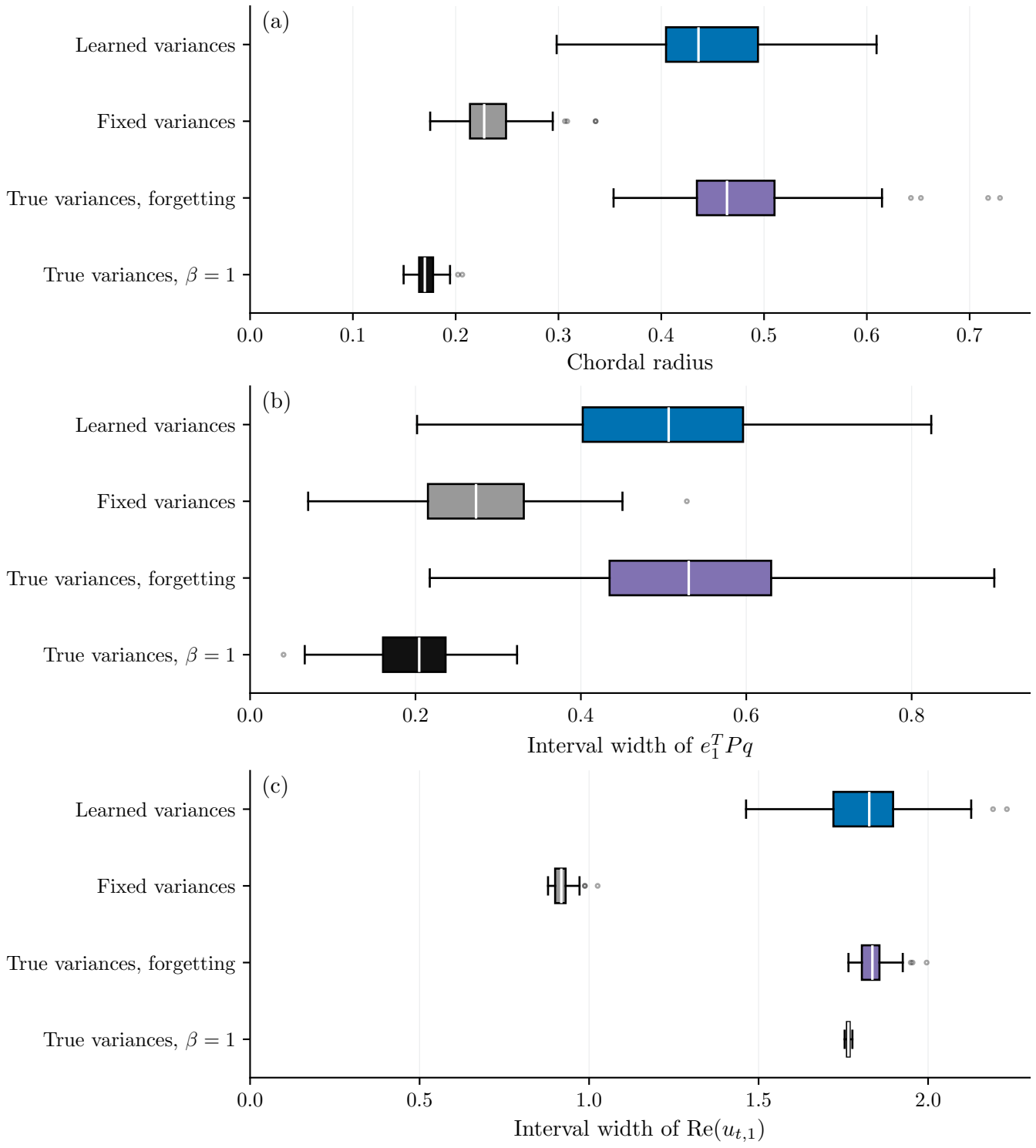

(a)
(b)
(c)
Observed inclusion
Nominal probability
0.0
0.2
0.4
0.6
0.8
1.0
0.50
0.80
0.95
True variances, $\beta = 1$
True variances, forgetting
Fixed variances
Learned variances

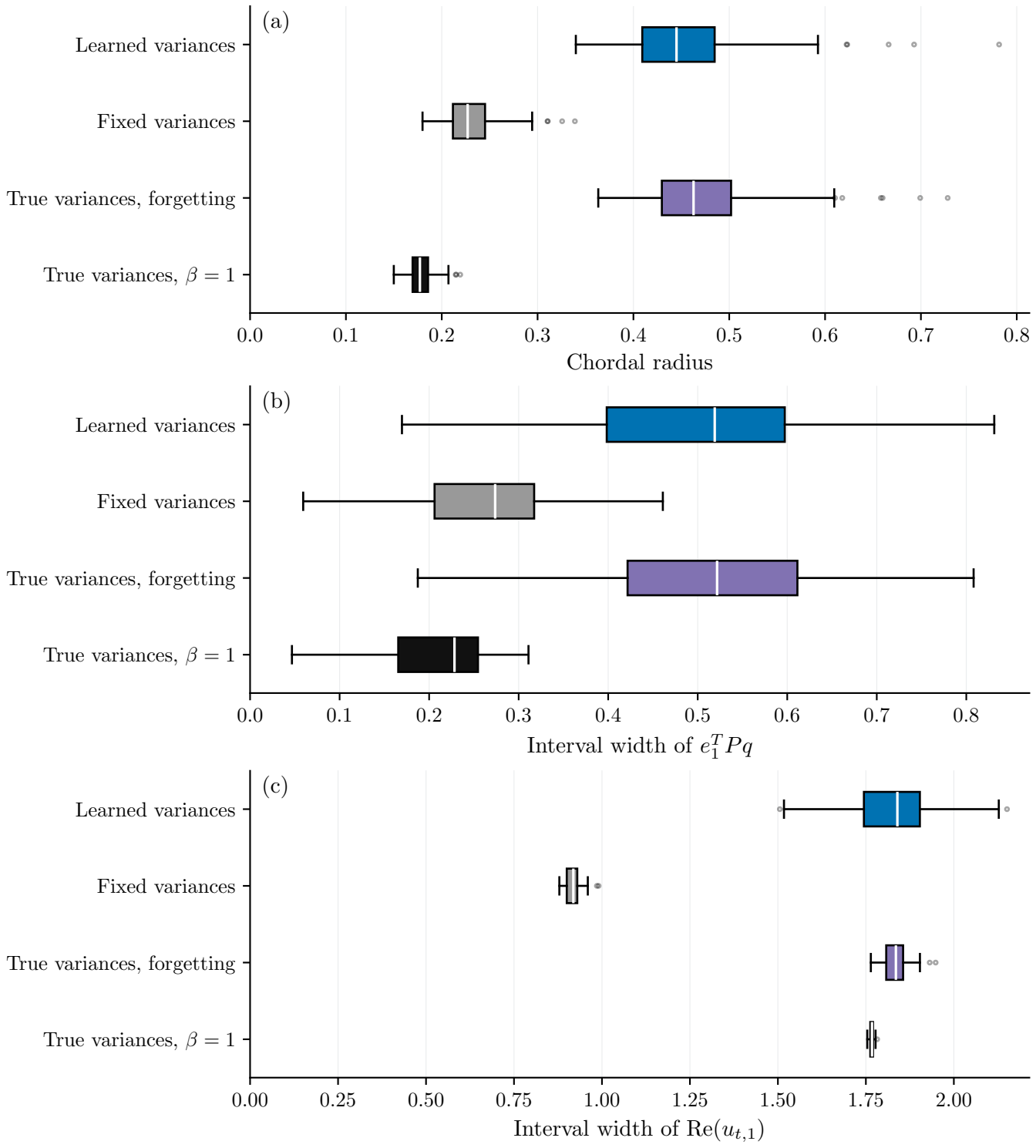

(a)
(b)
(c)
Observed inclusion
Nominal probability
0.0
0.2
0.4
0.6
0.8
1.0
0.50
0.80
0.95
True variances, $\beta = 1$
True variances, forgetting
Fixed variances
Learned variances

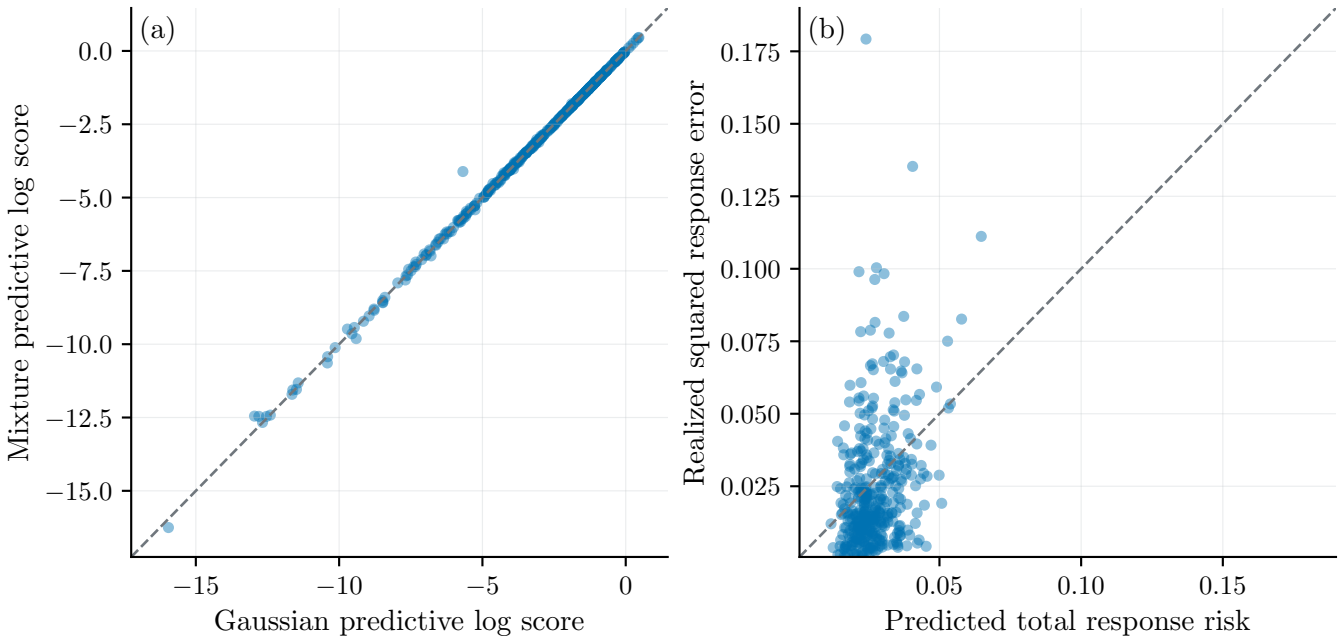

(a)
(b)
(c)
Observed inclusion
Nominal probability
0.0
0.2
0.4
0.6
0.8
1.0
0.50
0.80
0.95
True variances, $\beta = 1$
True variances, forgetting
Fixed variances
Learned variances

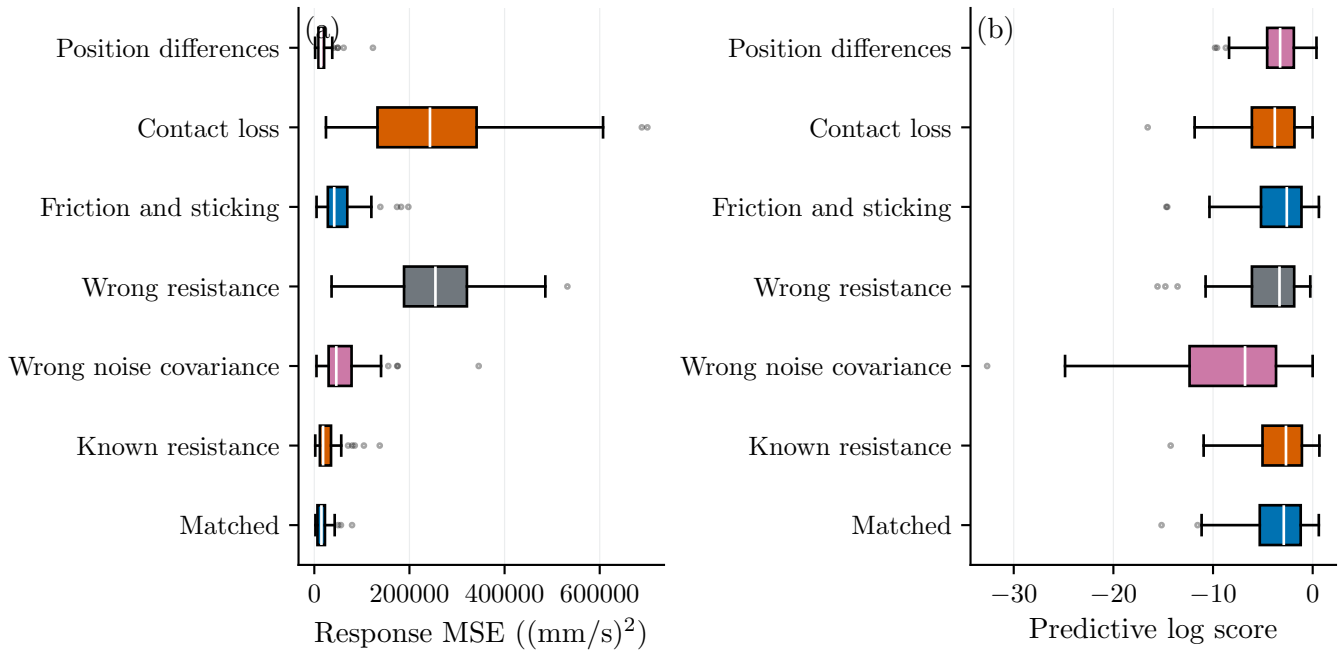

(a)
(b)
(c)
Observed inclusion
Nominal probability
0.0
0.2
0.4
0.6
0.8
1.0
0.50
0.80
0.95
True variances, $\beta = 1$
True variances, forgetting
Fixed variances
Learned variances

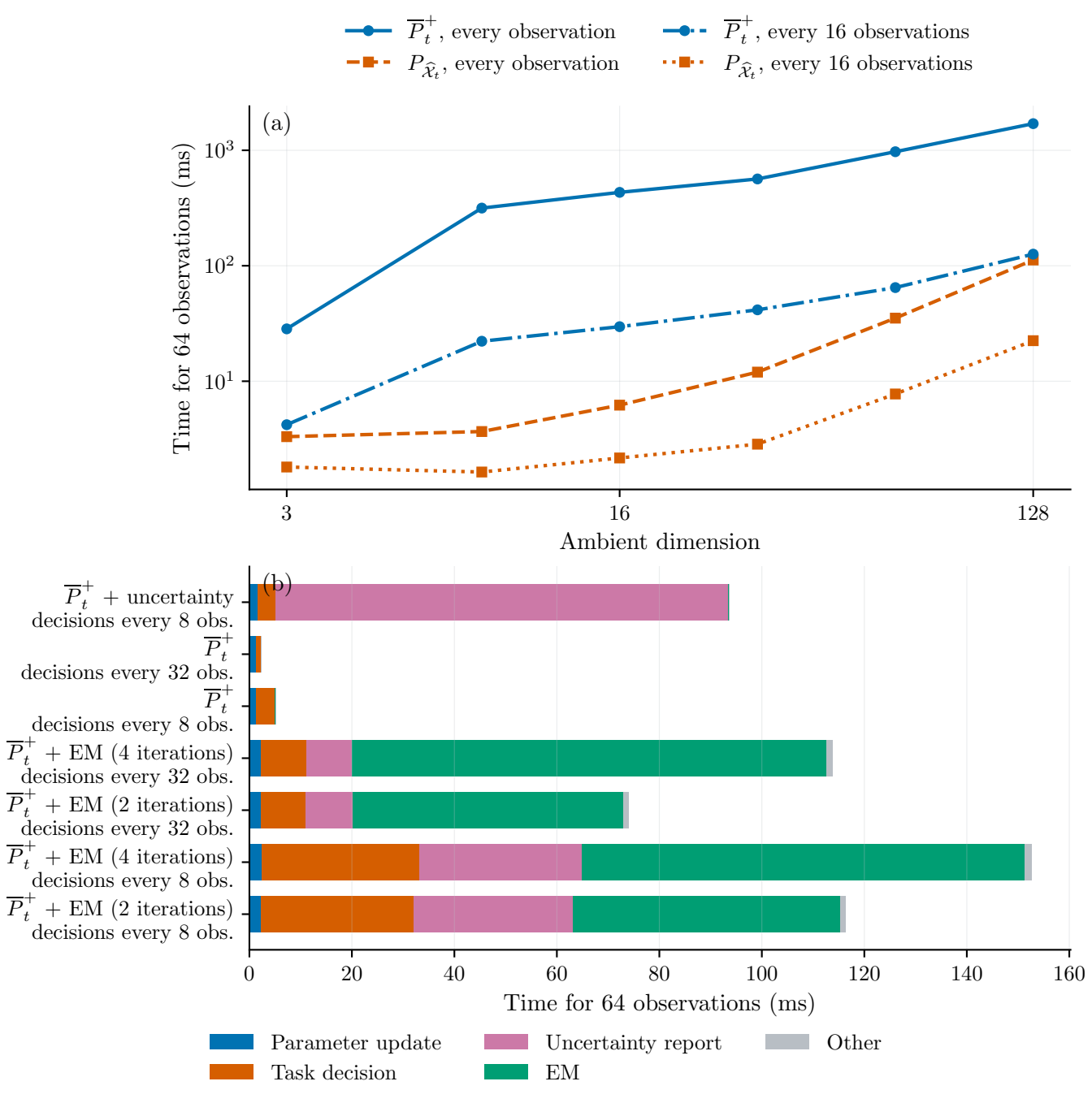

(a)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Chordal radius
(b)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Interval width of $e_1^T Pq$
(c)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.0
0.5
1.0
1.5
2.0
Interval width of $\mathrm{Re}(u_{t,1})$

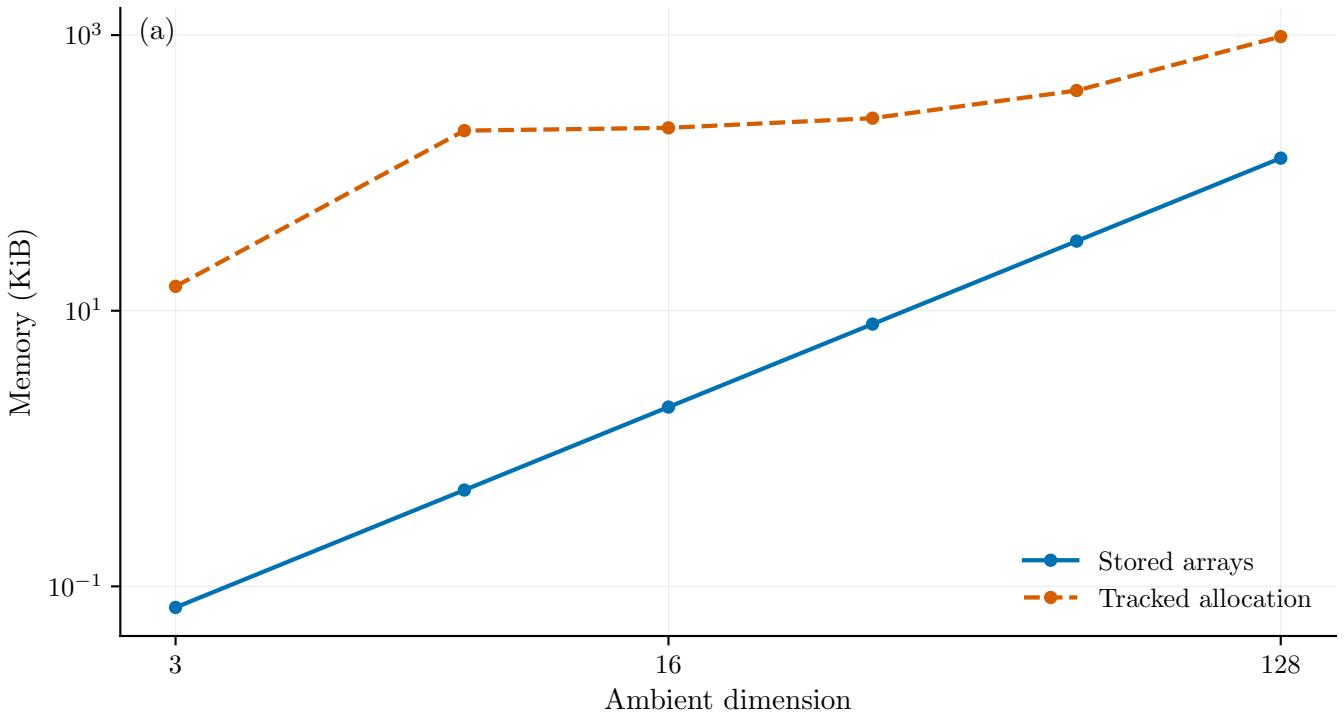

(a)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Chordal radius
(b)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.0
0.2
0.4
0.6
0.8
Interval width of $e_1^T P q$
(c)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.0
0.5
1.0
1.5
2.0
Interval width of $\mathrm{Re}(u_{t,1})$

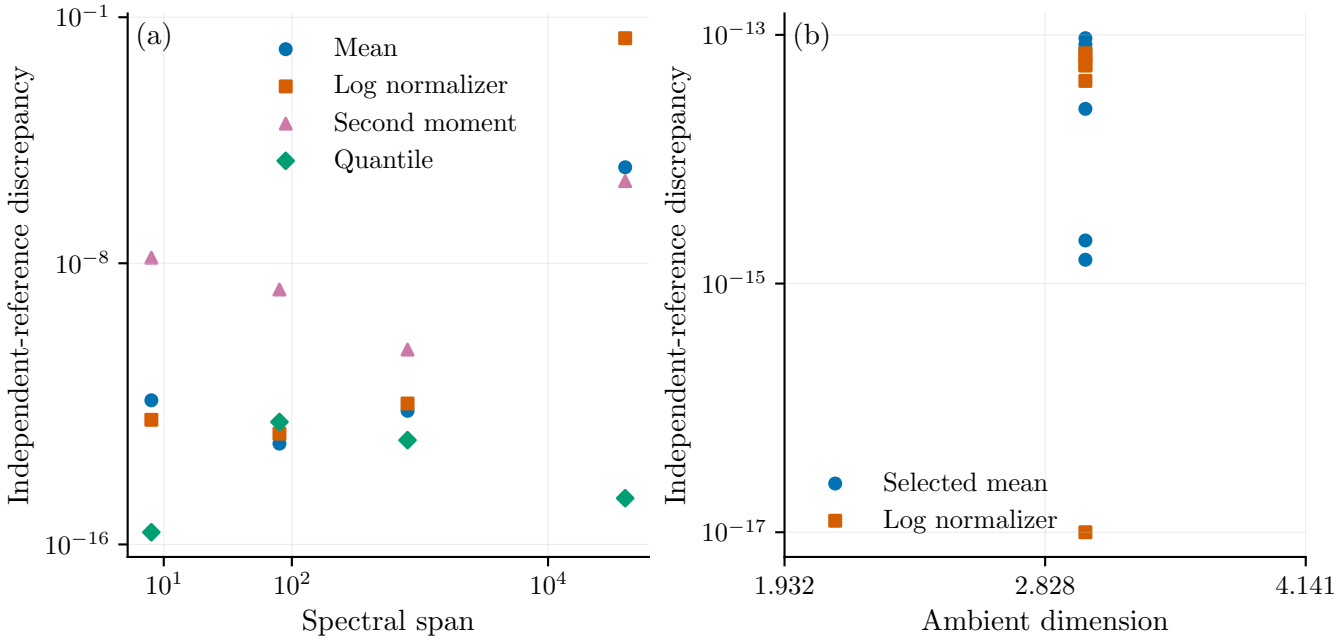

(a)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Chordal radius
(b)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Interval width of $e_1^T Pq$
(c)
Learned variances
Fixed variances
True variances, forgetting
True variances, $\beta = 1$
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Interval width of $\mathrm{Re}(u_{t,1})$

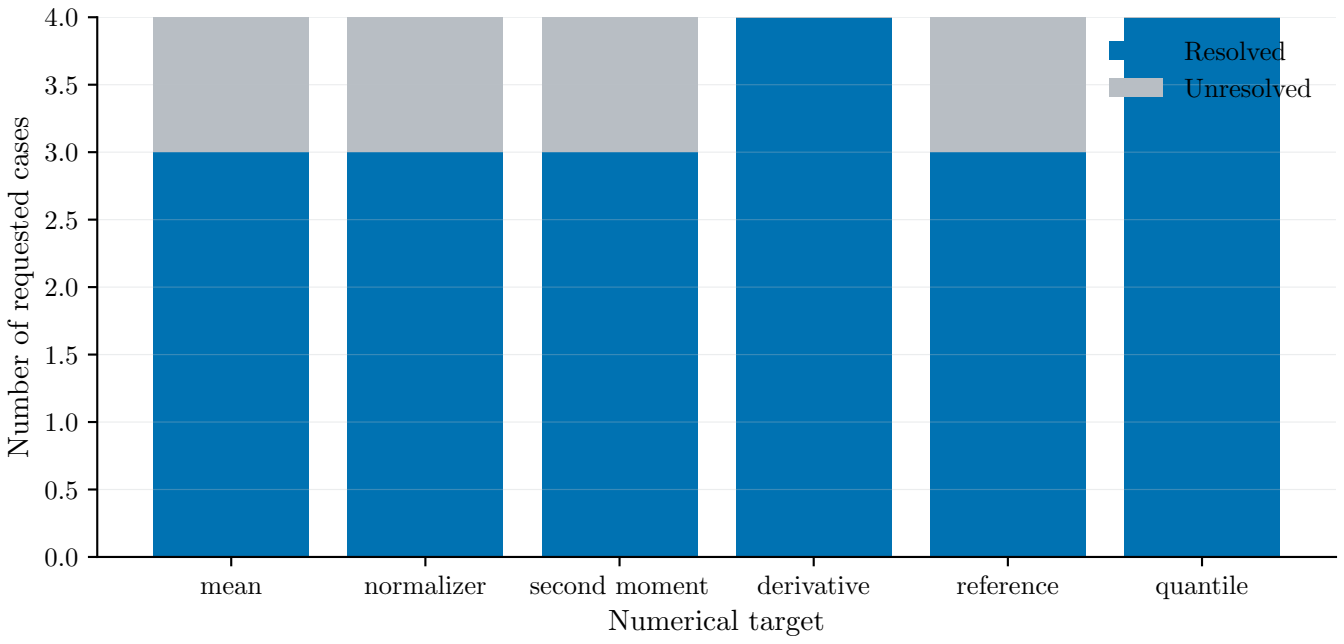

(a)
Mixture predictive log score
Gaussian predictive log score
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−15
−10
−5
0
(b)
Realized squared response error
Predicted total response risk
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.05
0.10
0.15

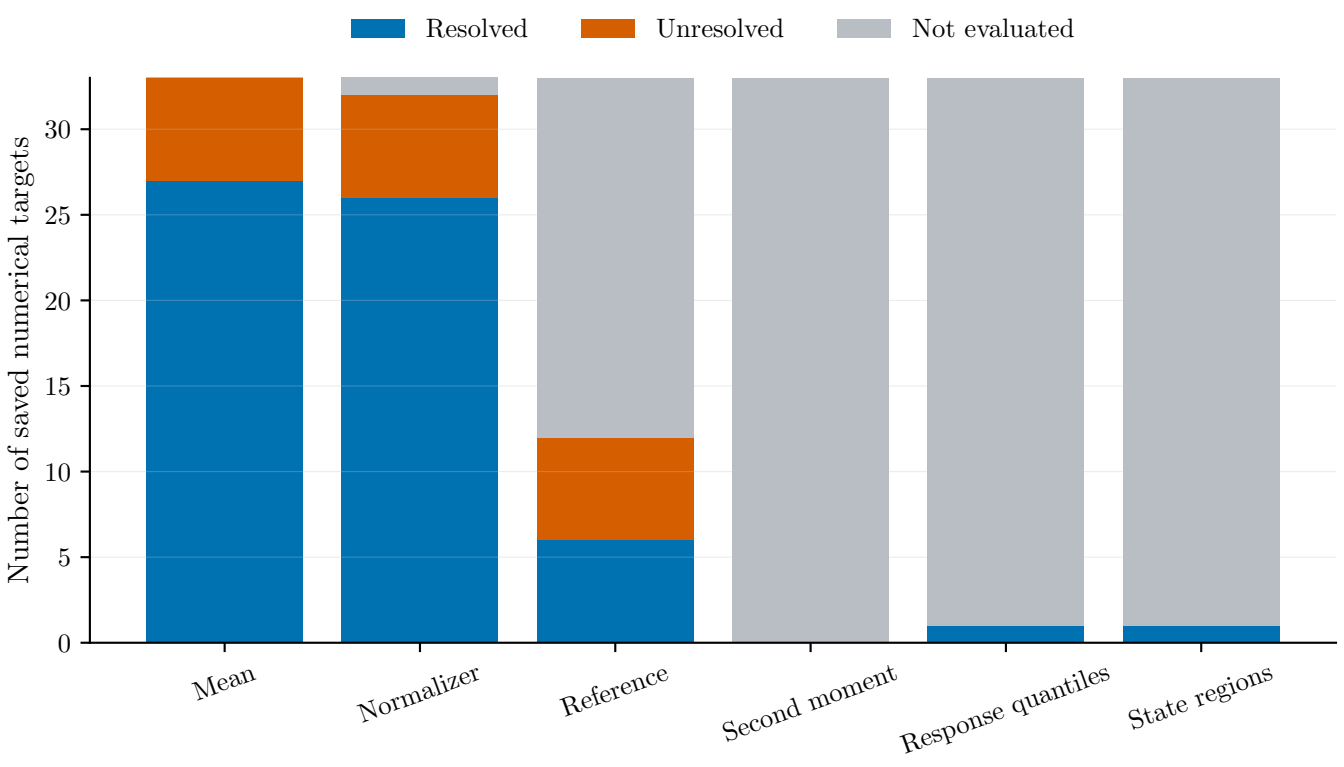

(a)
Position differences
Contact loss
Friction and sticking
Wrong resistance
Wrong noise covariance
Known resistance
Matched
0
200000
400000
600000
Response MSE ((mm/s)^2)
(b)
Position differences
Contact loss
Friction and sticking
Wrong resistance
Wrong noise covariance
Known resistance
Matched
−30
−20
−10
0
Predictive log score


$\overline{P}_t^+$, every observation
$\overline{P}_t^+$, every 16 observations
$P_{\widehat{\mathcal{X}}_t}$, every observation
$P_{\widehat{\mathcal{X}}_t}$, every 16 observations
(a)
Time for 64 observations (ms)
10^3
10^2
10^1
3
16
128
Ambient dimension
(b)
$\overline{P}_t^+$ + uncertainty decisions every 8 obs.
$\overline{P}_t^+$ decisions every 32 obs.
$\overline{P}_t^+$ decisions every 8 obs.
$\overline{P}_t^+$ + EM (4 iterations) decisions every 32 obs.
$\overline{P}_t^+$ + EM (2 iterations) decisions every 32 obs.
$\overline{P}_t^+$ + EM (4 iterations) decisions every 8 obs.
$\overline{P}_t^+$ + EM (2 iterations) decisions every 8 obs.
0
20
40
60
80
100
120
140
160
Time for 64 observations (ms)
Parameter update
Task decision
Uncertainty report
EM
Other


(a)
10^3
10^1
10^{-1}
Memory (KiB)
3
16
128
Ambient dimension
Stored arrays
Tracked allocation


(a)
Mean
Log normalizer
Second moment
Quantile
Independent-reference discrepancy
10^{-1}
10^{-8}
10^{-16}
10^{1}
10^{2}
10^{4}
Spectral span
(b)
Selected mean
Log normalizer
Independent-reference discrepancy
10^{-13}
10^{-15}
10^{-17}
1.932
2.828
4.141
Ambient dimension


Resolved
Unresolved
Number of requested cases
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
mean
normalizer
second moment
derivative
reference
quantile
Numerical target


Resolved
Unresolved
Not evaluated
Number of saved numerical targets
0
5
10
15
20
25
30
Mean
Normalizer
Reference
Second moment
Response quantiles
State regions